

Pioneer Project Accounting System

Android Mobile Rollout Plan

Phased implementation guide with screen-by-screen checklist

Version: 1.0 | **Date:** June 2026

Stack: Django 5.2 - Server-rendered HTML/JS

1. Executive Summary

The Pioneer Project Accounting System is a Django web application, not a native Android app. Recommended path: deploy securely, make priority screens mobile-responsive, enforce role-based access on the server, then add PWA install support.

Minimum viable rollout: Phases 0-1-2-3 (8-10 weeks). **Polished field version:** Phases 4-5 (12-14 weeks).

Android v1 focus: Login, Dashboard, Misc Purchase (partial buy + officer voucher), Budget Actuals (read), PDF/share for vouchers. Procurement and master data setup remain desktop-first.

2. How to Read This Document

Dimension	Meaning
Mobile UI	Task completable on phone without constant zoom/scroll
Backend guard	Access enforced in view, not just hidden in HTML
Print/PDF	Works on Android without desktop printer dialog
Network	Tolerates slow/interrupted mobile data
Priority	P0 = must-have for field Android v1

Readiness: **Not ready** | **Partial** | **Ready** | Desktop-only (intentional)

3. Phase 0 - Foundation (1-2 weeks)

Goal: App reachable and safe on mobile before UI changes.

Task	Why
Deploy Django to production	Phones need stable URL
Enable HTTPS	Secure login and PWA
Domain/IP + firewall	Staff bookmark one address
DB backups, error logging	Accounting cannot fail silently
Session timeout, strong passwords	Phones are easily lost
Decide mobile user groups	Drives screen priorities

Critical gap: login.html template is missing from /templates/. Mobile rollout cannot start until login page exists.

Exit test: Android Chrome -> https://your-domain/login/ -> sign in -> dashboard over HTTPS.

4. Phase 1 - Mobile Core Screens (3-5 weeks)

Priority order: Login, Dashboard, Misc Purchase + Partial Purchase, Officer Voucher, Misc Budget Actuals.

Change	Applies to
Single-column layout	All priority pages
44px+ tap targets	Purchase + voucher
Sticky action bar	Modals/forms
Cards instead of wide tables	RO lines, vouchers
Full-screen modals on phone	Partial purchase
Real device testing	Everything

De-prioritize: Setup CRUD, heavy dashboards, bulk master data.

5. Phase 2 - Access Control (2-3 weeks)

Define roles (GM, Officer, Finance, Admin). Map role to actions. Pass flags to templates. Block unauthorized POST/API. Restrict setup to Admin.

Today: Mostly @login_required only; has_module_access() exists but is not wired.

6. Phase 3 - PWA (1-2 weeks)

manifest.json, icons, light service worker, install prompt. No full offline accounting in v1.

7. Phase 4 - Documents (1-2 weeks)

Download PDF + Share instead of print dialog. Apply to officer voucher, MPO print, payment vouchers.

8. Phase 5 - Field Polish (2-3 weeks)

Deep links, session keep-alive, toasts, recent tasks, loading states on slow networks.

9. Phase 6 - Hybrid APK (optional, 4-6 weeks)

Capacitor wrapper to hosted URL. Branded APK / Play Store if needed.

10. Phase 7 - Native App (optional, 3-6+ months)

REST API + Flutter/React Native. Year 2 decision only.

11. Screen-by-Screen Checklist

Tier A - Field Operations (v1 Core)

Screen	URL	Priority	Mobile	Backend
Login	/login/	P0	Missing template	Auth only
Dashboard	/dashboard/	P0	Partial	Login only
Misc Purchase	/misc-purchase/	P0	Partial	Login only
GM Memo	/ad-hoc-memo/	P1	Desktop	No GM check
Authorize MPO	/authorize-mpo/	P1	N/A	No GM check
Print MPO	/print-mpo/id/	P1	Print only	Login only
Officer Voucher	/adhoc-officer-voucher/print/id/	P0	Partial	Login only
Budget Actuals	/misc-budget-actuals/	P0	No viewport	Login only
GM Disbursement	/gm-disbursement/	P2	Partial	Login only

Misc Purchase sub-areas

Area	Today	v1 Target
280px sidebar	Blocker	Drawer / bottom nav
RO lines table	Wide	Card per line
Partial purchase modal	Partial	Full-screen wizard
Buy now qty	Small	Large numeric inputs
Deep links	Exists	Must work on phone

Tier B - Financial Ops

Screen	URL	Priority	v1
Ops Dashboard	/ops-dashboard/	P2	Desktop recommended
Budget Overview	/budget-actuals/	P2	Read-only KPI
Budget Approval	/budget-approval/	P3	Defer

Tier C - Procurement (Desktop-first)

Screen	URL	Priority
BOM Builder	/bom-builder/	P3
RO Builder	/ro-builder/	P3
RFQ Manager	/rfq-manager/	P3
Bid Evaluation	/bid-evaluation/	P3
LPO Dispatch	/lpo-dispatch/	P3
LPO Settlement	/procurement/lpo-settlement/	P3

Tier D - Print Screens

Standard: phone preview + Download PDF + Share + desktop print.

Screen	URL	Priority
RO Print	/ro-print/id/	P2
Fluid RO	/print-fluid-ro/	P2
LPO Print	/lpo/print/id/	P3
GRN Print	/grn/print/id/	P3
Payment Voucher	/payment-voucher/print/id/	P2

Tier E - Admin (Desktop only)

User, Role, Bank, Supplier, GL, Task, Product CRUD - block non-admin on server.

12. Role and Access Matrix

Role	Misc Purchase	Voucher	Authorize	Actuals	Disburse	Setup	Procurement
Field Officer	Yes	Raise	No	Read	No	No	No
GM	Yes	Yes	Yes	Yes	Yes	No	Read
Finance	Read	Read	Read	Yes	Read	No	Read
Procurement	Read	Read	No	Read	No	No	Yes
Admin	Yes	Yes	Yes	Yes	Yes	Yes	Yes

13. UAT Scripts

UAT-1 Officer Partial Purchase (P0)

1. Login as Officer
2. Open /misc-purchase/?task_id=XXX
3. RO Listing and Partial Purchase
4. Tick lines, partial qty
5. Raise voucher
6. Save/share PDF

Pass: Under 5 min, no desktop fallback.

UAT-2 GM Budget (P0)

1. Login as GM
2. /misc-budget-actuals/?task_id=XXX
3. Read budget position
4. Open voucher from audit

UAT-3 GM Authorize (P1)

1. GM -> /ad-hoc-memo/ -> authorize -> print MPO PDF

UAT-4 Security (P0)

1. Officer tries /users/create/, /authorize-mpo/ directly

2. Must get 403

UAT-5 Deep Link (P0)

1. Open misc-purchase/?open_ro_purchase_list=1&ro_detail=5 on phone
2. Correct RO opens after login

14. Sprint Plan

Sprint	Weeks	Work	Outcome
S1	1-2	Login, HTTPS, dashboard nav	Sign in and navigate
S2	3-4	Misc purchase + voucher	Core field workflow
S3	5	PDF download/share	Documents on phone
S4	6	Budget actuals mobile	GM oversight
S5	7-8	Role enforcement	Safe rollout
S6	9	PWA	Home-screen app
S7	10+	GM memo, polish	Broader mobile

Success metrics

- Mobile login success $\geq 95\%$
- Partial purchase to voucher without desktop
- No unauthorized URL/API access
- 80%+ staff install PWA
- Fewer phone support tickets after Phase 1

15. Summary Scorecard

Area	Today	Phase 1	Phase 3-5
Login	Not ready	Ready	Ready
Dashboard	Partial	Ready	Ready
Misc Purchase	Partial	Ready	Ready
Officer Voucher	Partial	Ready	Ready
Budget Actuals	Not ready	Ready	Ready
GM Memo	Not ready	Partial	Ready
Procurement chain	Not ready	Desktop	Selective
Access control	Not ready	Partial	Ready
PWA	Not ready	Not ready	Ready

16. Decisions Before Build

1. **Who gets Android v1?** Officers only or GM too?
2. **GM authorization on phone?** If yes, memo moves to P0.
3. **Distribution:** Browser/PWA only or branded APK?

Recommended order

1. Phase 0 deploy
2. Phase 1 misc purchase + voucher
3. Phase 2 access control
4. Phase 3 PWA
5. Phase 4 PDF/share
6. Phase 5 polish
7. Phase 6 APK only if needed

Planning guidance for Pioneer Project Accounting System. Estimates assume dedicated dev, IT support, and UAT on physical Android devices.